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United States Patent	12418980
Kind Code	B2
Date of Patent	September 16, 2025
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Wiring board

Abstract

A wiring board that includes: a wiring conductor; a first dielectric layer around the wiring conductor and containing a first glass and a first ceramic filler; and a second dielectric layer interposed between the wiring conductor and the first dielectric layer, the second dielectric layer being in contact with the wiring conductor and the first dielectric layer, and the second dielectric layer containing a second glass and a second ceramic filler. A sintering temperature of the second glass contained in the second dielectric layer is higher than a sintering temperature of the wiring conductor, and a grain size of the second glass contained in the second dielectric layer is smaller than a grain size of the first glass contained in the first dielectric layer.

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Appl. No.: 18/485650

Filed: October 12, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240040694 A1	Feb. 01, 2024

Foreign Application Priority Data

JP	2021-072808	Apr. 22, 2021
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Related U.S. Application Data

Publication Classification

Int. Cl.: **H05K1/03** (20060101); **C03C4/16** (20060101); **C03C14/00** (20060101); **H05K1/11** (20060101)

U.S. Cl.:

CPC **H05K1/0306** (20130101); **C03C4/16** (20130101); **C03C14/004** (20130101); **H05K1/115** (20130101); C03C2204/00 (20130101); C03C2214/04 (20130101); C03C2214/20 (20130101); H05K2201/0175 (20130101); H05K2201/0209 (20130101)

Field of Classification Search

CPC: H05K (1/0306); H05K (1/115); H05K (1/024); H05K (2201/0175); H05K (2201/0209); H05K (2201/017); H05K (2201/0187); H05K (2201/0195); C03C (4/16); C03C (4/004); C03C (2204/00); C03C (2214/04); C03C (2214/20); C03C (10/00); C03C (17/3607); C03C (17/3649); H01L (23/15)

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation of International application No. PCT/JP2022/003274, filed Jan. 28, 2022, which claims priority to Japanese Patent Application No. 2021-072808, filed Apr. 22, 2021, the entire contents of each of which are incorporated herein by reference.

TECHNICAL FIELD

(1) The present disclosure relates to a wiring board including a wiring conductor and a dielectric layer adjacent to the wiring conductor and containing a glass and a ceramic filler.

BACKGROUND ART

(2) Glass-ceramics contain a crystallized glass and a ceramic filler. A wiring board in which glass-ceramics are used for a dielectric layer has high strength and excellent heat resistance because of the crystallized glass.

(3) On the other hand, in a wiring board in which glass-ceramics are used for a dielectric layer, crystals are precipitated from the glass, and thus, irregularities are generated at an interface between the dielectric layer and a wiring conductor, resulting in roughness. When an electric signal passing through the wiring conductor has a high frequency, the electric signal passing through the wiring conductor concentrates near the interface with the dielectric layer because of a skin effect. Thus, when the interface between the dielectric layer and the wiring conductor is rough, the conductivity of the wiring conductor decreases, and transmission loss may occur.

(4) In the wiring board of Patent Document 1, an intermediate layer made of a non-crystallized glass is interposed between a ceramic insulating layer corresponding to the dielectric layer and a wiring conductor. Thus, in the wiring board of Patent Document 1, crystals are not deposited from glass in the intermediate layer. Therefore, the roughness of the interface between the wiring conductor and the adjacent material can be reduced as compared with a structure without the intermediate layer. Patent Document 1: JP-B2-4703212

SUMMARY OF THE DISCLOSURE

(5) However, the wiring board of Patent Document 1 reduces the roughness of the interface with the wiring conductor by not depositing crystals from glass. Thus, in the wiring board of Patent Document 1, only a structure in which a non-crystallized glass is disposed at the interface with the wiring conductor can be adopted, and thus there is room for further improvement.

(6) An object of the present disclosure is to solve the above problems and to provide a wiring board capable of reducing the roughness of an interface between a dielectric layer and a wiring conductor as compared with the conventional technology while reducing the limitation of the structure.

(7) To achieve the above object, the present disclosure is configured as follows. A wiring board according to one aspect of the present disclosure includes: a wiring conductor; a first dielectric layer around the wiring conductor and containing a first glass and a first ceramic filler; and a second dielectric layer interposed between the wiring conductor and the first dielectric layer, the second dielectric layer being in contact with the wiring conductor and the first dielectric layer, and the second dielectric layer containing a second glass and a second ceramic filler, wherein a sintering temperature of the second glass contained in the second dielectric layer is higher than a sintering temperature of the wiring conductor, and a grain size of the second glass contained in the second dielectric layer is smaller than a grain size of the first glass contained in the first dielectric layer.

(8) According to the present disclosure, it is possible to reduce the roughness of the interface between the dielectric layer and the wiring conductor as compared with the conventional technology while reducing the limitation of the structure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a plan view of a wiring board according to a first embodiment of the present disclosure.
- (2) FIG. 2 is a longitudinal sectional view showing a section taken along the line A-A in FIG. 1.
- (3) FIG. 3 is a longitudinal sectional view of a green sheet on which a ceramic paste is stacked.
- (4) FIG. 4 is a longitudinal sectional view of a green sheet in which a conductor paste is stacked on the ceramic paste of FIG. 3.
- (5) FIG. 5 is a longitudinal sectional view of a green sheet in which a ceramic paste is stacked on the ceramic paste and the conductor paste of FIG. 4.
- (6) FIG. 6 is a longitudinal sectional view showing stacking of another green sheet on the green sheet of FIG. 5.
- (7) FIG. 7 is a table showing results of a performance evaluation test.
- (8) FIG. 8 is a longitudinal sectional view of a wiring board according to a second embodiment of the present disclosure, corresponding to the section taken along the line A-A in FIG. 1.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

- (9) A wiring board according to one aspect of the present disclosure includes: a wiring conductor; a first dielectric layer around the wiring conductor and containing a first glass and a first ceramic filler; and a second dielectric layer interposed between the wiring conductor and the first dielectric layer, the second dielectric layer being in contact with the wiring conductor and the first dielectric layer, and the second dielectric layer containing a second glass and a second ceramic filler, wherein a sintering temperature of the second glass contained in the second dielectric layer is higher than a sintering temperature of the wiring conductor, and a grain size of the second glass contained in the second dielectric layer is smaller than a grain size of the first glass contained in the first dielectric layer.
- (10) According to this configuration, the grain size of the second glass contained in the second dielectric layer is smaller than the grain size of the first glass contained in the first dielectric layer. Thus, the roughness of the interface between the second dielectric layer and the wiring conductor can be reduced as compared with a configuration in which the second dielectric layer is not provided and the first dielectric layer is in contact with the wiring conductor. As a result, transmission loss in the wiring conductor can be reduced.
- (11) According to this configuration, the sintering temperature of the second glass contained in the second dielectric layer is higher than the sintering temperature of the wiring conductor. Thus, during sintering of the wiring board, the wiring conductor is sintered before the second glass contained in the second dielectric layer is sintered in the process of increasing the temperature. This causes the second glass contained in the second dielectric layer to maintain the original grain shape during sintering of the wiring conductor. As a result, the roughness of the interface between the wiring conductor and the second dielectric layer can depend on the grain size of the second glass contained in the second dielectric layer.
- (12) If the grain size of the first glass contained in the first dielectric layer is as small as the grain size of the second glass contained in the second dielectric layer, that is, if the glass of all the dielectric layers is atomized, the degreasing property at the time of sintering the wiring board deteriorates. As a result, the gas generated by the combustion of a binder is less likely to escape, and the possibility of occurrence of structural defects such as peeling of the dielectric layer from the wiring conductor increases. According to this configuration, the grain size of the second glass contained in the second dielectric layer is smaller than the grain size of the first glass contained in the first dielectric layer. That is, only the glass of some dielectric layers is atomized. This can reduce occurrence of structural defects such as peeling of the dielectric layer from other layers as compared with a configuration in which the glass of all the dielectric layers is atomized.
- (13) According to this configuration, the first glass contained in the first dielectric layer and the

second glass contained in the second dielectric layer may be a crystallized glass or a non-crystallized glass. That is, according to this configuration, the restriction of the structure of the wiring board can be reduced.

(14) In the wiring board, the grain size of the second ceramic filler contained in the second dielectric layer may be smaller than the grain size of the first ceramic filler contained in the first dielectric layer.

(15) According to this configuration, the grain size of the second ceramic filler contained in the second dielectric layer is smaller than the grain size of the first ceramic filler contained in the first dielectric layer. Thus, the roughness of the interface between the dielectric layer and the wiring conductor can be reduced as compared with a configuration in which the second dielectric layer is not provided and the first dielectric layer is in contact with the wiring conductor.

(16) In the wiring board, a through via that penetrates the first dielectric layer and the second dielectric layer and is electrically connected to the wiring conductor may be formed in the first dielectric layer and the second dielectric layer.

(17) According to this configuration, in the vicinity of the boundary between the through via and the wiring conductor, a decrease in conductivity of the wiring conductor is prevented because of a reduction in roughness of the interface between the wiring conductor and the second dielectric layer. Thus, since the roughness of the interface between a part of the through via and the second dielectric layer is reduced, transmission loss between the wiring conductor and the through via can be reduced as compared with a configuration in which only the first dielectric layer is provided.

(18) In the wiring board, the through via may have a tapered shape that reduces its diameter in a direction toward the wiring conductor.

(19) Normally, in a through via having a tapered shape, transmission loss is likely to occur in a part having a smaller diameter than a part having a larger diameter. According to this configuration, the part having a smaller diameter of the through via where transmission loss is likely to occur is positioned near the boundary with the wiring conductor. In the vicinity of the boundary, a decrease in conductivity of the wiring conductor is prevented because of a reduction in roughness of the interface between the wiring conductor and the second dielectric layer. Thus, transmission loss between the wiring conductor and the through via can be reduced.

First Embodiment

(20) FIG. 1 is a plan view of a wiring board according to a first embodiment of the present disclosure. FIG. 2 is a longitudinal sectional view showing a section taken along the line A-A in FIG. 1. In the wiring board, a wiring conductor is formed adjacent to a dielectric layer containing a crystallized glass and a ceramic filler. In the wiring board according to the first embodiment, a wiring conductor is formed inside.

(21) As shown in FIGS. 1 and 2, a wiring board **10** has a rectangular parallelepiped shape. The shape of the wiring board **10** is not limited to a rectangular parallelepiped shape.

(22) As shown in FIG. 2, the wiring board **10** includes a wiring conductor **20** and dielectric layers **30** and **40**. The dielectric layer **30** is an example of the second dielectric layer. The dielectric layer **40** is an example of the first dielectric layer.

(23) The wiring conductor **20** functions as a wiring that electrically connects a plurality of electronic components (not shown) to be mounted on the wiring board **10** to each other. The wiring conductor **20** can also function as an electrode to which an electronic component to be mounted on the wiring board **10** is electrically connected via solder or the like.

(24) In the first embodiment, the wiring conductor **20** is covered with the dielectric layer **30**. The wiring conductor **20** is formed inside the wiring board **10**.

(25) The wiring conductor **20** is obtained by printing a conductive paste on a surface of the dielectric layer **30**. The conductive paste is made of, for example, copper.

(26) The dielectric layer **30** covers the periphery of the wiring conductor **20**. The dielectric layer **30** is in contact with the wiring conductor **20**.

(27) The dielectric layer **30** is made of a glass-ceramic. The dielectric layer **30** contains a crystallized glass and a ceramic filler. In the first embodiment, the dielectric layer is made of a crystallized glass and alumina and zirconia constituting the ceramic filler. In the first embodiment, the proportion of each material contained in the dielectric layer **30** is as follows. That is, the crystallized glass contained in the dielectric layer **30** is 98 (vol %) of the dielectric layer **30**. The alumina contained in the dielectric layer **30** is 1.5 (vol %) of the dielectric layer **30**. The zirconia contained in the dielectric layer **30** is 0.5 (vol %) of the dielectric layer **30**.

(28) The dielectric layer **40** covers the periphery of the dielectric layer **30**. The dielectric layer **40** is in contact with the dielectric layer **30**. On the other hand, in the first embodiment, the dielectric layer **40** is not in contact with the wiring conductor **20**. That is, the dielectric layer **40** is provided around the wiring conductor **20**, and the dielectric layer **30** is interposed between the wiring conductor **20** and the dielectric layer **40**.

(29) The dielectric layer **40** is made of a glass-ceramic. The dielectric layer **40** contains a crystallized glass and a ceramic filler. In the first embodiment, the dielectric layer contains the same materials as the dielectric layer **30** at the same proportions as in the dielectric layer **30**. That is, dielectric layer **40** is made of a crystallized glass and alumina and zirconia constituting the ceramic filler. In the first embodiment, the crystallized glass contained in the dielectric layer **40** is 98 (vol %) of the dielectric layer **30**. The alumina contained in the dielectric layer **40** is 1.5 (vol %) of the dielectric layer **30**. The zirconia contained in the dielectric layer **40** is 0.5 (vol %) of the dielectric layer **30**.

(30) The grain size of the crystallized glass (specifically, the glass powder constituting the crystallized glass) contained in the dielectric layer **30** is smaller than the grain size of the crystallized glass contained in the dielectric layer **40**. In the first embodiment, the grain size of each material (crystallized glass, alumina, and zirconia) contained in the dielectric layer **30** is the average grain size in the whole of the dielectric layer **30** or in any part of the dielectric layer **30** when the part is cut out. In the same manner, the grain size of each material (crystallized glass, alumina, and zirconia) contained in the dielectric layer **40** is the average grain size in the whole of the dielectric layer **40** or in any part of the dielectric layer **40** when the part is cut out.

(31) The grain size of the crystallized glass contained in the dielectric layer **30** is smaller than the grain size of the crystallized glass contained in the dielectric layer **40**. In the first embodiment, the grain size of the crystallized glass contained in the dielectric layer is 2.0 (μm) or less. In the first embodiment, the grain size of the crystallized glass contained in the dielectric layer **30** is 1.0 (μm) or less.

(32) The grain size of each ceramic filler (specifically, alumina and zirconia) contained in the dielectric layer **30** is equal to or smaller than the grain size of each ceramic filler (specifically, alumina and zirconia) contained in the dielectric layer **40**.

(33) In the first embodiment, the grain size of the alumina contained in the dielectric layer **40** is 2.5 (μm). The grain size of the alumina contained in the dielectric layer **30** is 2.5 (μm) or less.

(34) In the first embodiment, the grain size of the zirconia contained in the dielectric layer **40** is 1.3 (μm). The grain size of the zirconia contained in the dielectric layer **30** is 1.3 (μm) or less.

(35) The sintering temperature of the crystallized glass contained in the dielectric layer is higher than the sintering temperature of the wiring conductor **20**. In the first embodiment, the sintering temperature of the crystallized glass contained in the dielectric layer **30** is 900 to 950 ($^{\circ}\text{C}$.). The sintering temperature of the wiring conductor **20** is 850 to 900 ($^{\circ}\text{C}$.).

(36) In the first embodiment, a thickness T1 of the wiring conductor **20** is larger than a thickness T2 of the dielectric layer **30**. In the first embodiment, the thickness T1 of the wiring conductor **20** is 5 to 30 (μm), and the thickness T2 of the dielectric layer **30** is 5 to 20 (μm). The thickness T1 may be equal to or less than the thickness T2.

(37) Hereinafter, an example of a method for producing the wiring board **10** according to the first embodiment will be described. FIG. 3 is a longitudinal sectional view of a green sheet on which a

ceramic paste is stacked. FIG. 4 is a longitudinal sectional view of a green sheet in which a conductor paste is stacked on the ceramic paste of FIG. 3. FIG. 5 is a longitudinal sectional view of a green sheet in which a ceramic paste is stacked on the ceramic paste and the conductor paste of FIG. 4. FIG. 6 is a longitudinal sectional view showing stacking of another green sheet on the green sheet of FIG. 5.

(38) First, a green sheet **41** constituting a part of the dielectric layer **40** is produced by a known process (see FIG. 3). In the first embodiment, as described above, the materials of the green sheet **41** are 98 (vol %) of crystallized glass, 1.5 (vol %) of alumina, and 0.5 (vol %) of zirconia.

(39) Next, as shown in FIG. 3, a ceramic paste **31** constituting a part of the dielectric layer **30** is printed on a principal surface **41A** of the green sheet **41** by a known process. In the first embodiment, the materials of the ceramic paste **31** are the same as that of the green sheet **41**. However, the grain size of the crystallized glass contained in the ceramic paste **31** is smaller than the grain size of the crystallized glass contained in the green sheet **41**. The grain size of each of the alumina and zirconia contained in the ceramic paste **31** is equal to or smaller than the grain size of each of the alumina and zirconia contained in the green sheet **41**.

(40) Next, as shown in FIG. 4, a conductive paste **21** constituting the wiring conductor is printed on a principal surface **31A** of the ceramic paste **31** by a known process. In the first embodiment, the material of the conductive paste **21** is copper.

(41) Next, as shown in FIG. 5, a ceramic paste **32** constituting the rest of the dielectric layer **30** is printed on the principal surface **31A** of the ceramic paste **31** by a known process so as to cover the conductive paste **21**. This causes the conductive paste **21** to be buried in the dielectric layer **30** constituted by the ceramic pastes **31** and **32**. The material of the ceramic paste **32** is the same as that of the ceramic paste **31**.

(42) Next, as shown in FIG. 6, a green sheet **42** constituting the rest of the dielectric layer **40** is stacked on the principal surface **41A** of the green sheet **41** so as to cover the dielectric layer **30** by a known process. This causes the dielectric layer **30** to be buried in the dielectric layer **40** constituted by the green sheets **41** and **42**. The material of the green sheet **42** is the same as that of the green sheet **41**.

(43) Next, a stack formed by the step shown in FIG. 6 is sintered. The wiring board shown in FIG. 2 is thus completed.

(44) Hereinafter, results of a performance evaluation test performed on the wiring boards according to examples 1 to 6 and the wiring boards according to comparative examples 1 and 2 will be described. FIG. 7 is a table showing the results of the performance evaluation test.

(45) In each of the wiring boards **10** according to examples 1 to 3, the grain size of the crystallized glass contained in the dielectric layer **30** is smaller than the grain size of the crystallized glass contained in the dielectric layer **40**. On the other hand, in each of the wiring boards **10** according to examples 1 to 3, the grain size of each of the alumina and zirconia contained in the dielectric layer **30** is the same as the grain size of each of the alumina and zirconia contained in the dielectric layer **40**.

(46) In each of the wiring boards **10** according to examples 4 to 6, the grain size of the crystallized glass contained in the dielectric layer **30** is smaller than the grain size of the crystallized glass contained in the dielectric layer **40**. In each of the wiring boards **10** according to examples 4 to 6, the grain size of each of the alumina and zirconia contained in the dielectric layer **30** is smaller than the grain size of each of the alumina and zirconia contained in the dielectric layer **40**.

(47) Each of the wiring boards according to comparative examples 1 and 2 includes the wiring conductor **20** and the dielectric layers **30** and **40**, as in each of the wiring boards **10** according to examples 1 to 6. However, in each of the wiring boards according to comparative examples 1 and 2, the grain size of the crystallized glass contained in the dielectric layer **30** is the same as the grain size of the crystallized glass contained in the dielectric layer **40**. In each of the wiring boards according to comparative examples 1 and 2, the grain size of each of the alumina and zirconia

contained in the dielectric layer **30** is the same as the grain size of each of the alumina and zirconia contained in the dielectric layer **40**.

(48) As the performance evaluation, three items of Rmax, presence or absence of transmission loss, and presence or absence of delamination were evaluated.

(49) Rmax is the maximum height of the irregularities of the interface between the wiring conductor **20** and the dielectric layer **30**, and the unit is micrometers (μm).

(50) A tri-plate line having a width of 50 (μm) and a thickness of 7 (μm) of the wiring conductor **20** was produced, and the presence or absence of transmission loss was evaluated. When transmission is improved by 0.01 (dB/mm) or more at 60 (GHz), “○” is given in FIG. 7 as an evaluation of transmission loss. When transmission is improved by 0.02 (dB/mm) or more at 60 (GHz), double circles is given in FIG. 7 as an evaluation of transmission loss. When the improvement in transmission is less than 0.01 (dB/mm), it is assumed that transmission loss has occurred, and “x” is given in FIG. 7.

(51) The delamination is peeling of the dielectric layer **30** from the wiring conductor. When a section of the wiring board **10** is observed, and there is even one gap of 5 (μm) or more continuing 1 (mm) or more in a length direction in the section, it is assumed that delamination has occurred, and “x” is given in FIG. 7. In the other cases than the above, it is assumed that delamination has not occurred, and “○” is given in FIG. 7.

(52) As can be seen from FIG. 7, in the wiring board **10** according to example 3, the value of Rmax is smaller than that of the wiring boards according to comparative examples 1 and 2. That is, it can be seen that, in the wiring board **10** according to example 3, the irregularities of the interface between the wiring conductor **20** and the dielectric layer **30** is reduced as compared with the wiring boards according to comparative examples 1 and 2.

(53) As can be seen from FIG. 7, in the wiring boards **10** according to examples 1 to 3, transmission loss is practically acceptable, and delamination does not occur. On the other hand, transmission loss has occurred in the wiring board according to comparative example 1, and delamination has occurred in the wiring board according to comparative example 2.

(54) As can be seen from FIG. 7, in the wiring boards **10** according to examples 4 to 6, the value of Rmax is smaller than that of the wiring boards according to examples 1 to 3. That is, it can be seen that, in the wiring boards **10** according to examples 4 to 6, the irregularities of the interface between the wiring conductor **20** and the dielectric layer **30** is reduced as compared with the wiring boards according to examples 1 to 3.

(55) As can be seen from FIG. 7, in the wiring boards **10** according to examples 4 to 6, transmission loss is reduced as compared with the wiring boards **10** according to example 1 to 3. In the wiring boards **10** according to examples 4 to 6, delamination does not occur as in the wiring boards **10** according to examples 1 to 3.

(56) According to the first embodiment, the grain size of the crystallized glass contained in the dielectric layer **30** is smaller than the grain size of the crystallized glass contained in the dielectric layer **40**. Thus, the roughness of the interface between the dielectric layer **30** and the wiring conductor **20** can be reduced as compared with a configuration in which the dielectric layer **30** is not provided and the dielectric layer **40** is in contact with the wiring conductor **20**. As a result, transmission loss in the wiring conductor **20** can be reduced. In addition, peeling of the wiring conductor **20** from the dielectric layer can be reduced as compared with the conventional technology.

(57) According to the first embodiment, the sintering temperature of the crystallized glass contained in the dielectric layer **30** is higher than the sintering temperature of the wiring conductor **20**. Thus, during sintering of the wiring board **10**, the wiring conductor **20** is sintered before the crystallized glass contained in the dielectric layer **30** is sintered in the process of increasing the temperature. This causes the crystallized glass contained in the dielectric layer **30** to maintain the original grain shape during sintering of the wiring conductor **20**. As a result, the roughness of the

interface between the wiring conductor **20** and the dielectric layer **30** can depend on the grain size of the crystallized glass contained in the dielectric layer **30**.

(58) If the grain size of the crystallized glass contained in the dielectric layer **40** is as small as the grain size of the crystallized glass contained in the dielectric layer **30**, that is, if the glass of all the dielectric layers (dielectric layers **30** and **40**) is atomized, the degreasing property at the time of sintering the wiring board **10** deteriorates. As a result, the gas generated by the combustion of a binder is less likely to escape, and the possibility of occurrence of structural defects such as peeling of the dielectric layer from the wiring conductor increases. According to the first embodiment, the grain size of the crystallized glass contained in the dielectric layer **30** is smaller than the grain size of the crystallized glass contained in the dielectric layer **40**. That is, only the crystallized glass of some dielectric layers (dielectric layers **30**) is atomized. This can reduce occurrence of structural defects such as peeling of the dielectric layer from other layers as compared with a configuration in which the crystallized glass of all the dielectric layers (dielectric layers **30** and **40**) is atomized.

(59) According to the first embodiment, both the dielectric layers **30** and **40** contain a crystallized glass. Thus, the strength and heat resistance of the wiring board **10** can be favorably maintained as compared with a configuration in which a non-crystallized glass is used instead of a crystallized glass for at least one of the dielectric layers **30** and **40**.

(60) According to the first embodiment, the grain size of the ceramic filler contained in the dielectric layer **30** is smaller than the grain size of the ceramic filler contained in the dielectric layer **40**. Thus, the roughness of the interface between the dielectric layer **30** and the wiring conductor **20** can be reduced as compared with a configuration in which the dielectric layer **30** is not provided and the dielectric layer **40** is in contact with the wiring conductor **20**.

(61) In the first embodiment, the grain size of each ceramic filler (specifically, alumina and zirconia) contained in the dielectric layer **30** is equal to or smaller than the grain size of each ceramic filler (specifically, alumina and zirconia) contained in the dielectric layer **40**. However, the grain size of each ceramic filler contained in the dielectric layer **30** may be larger than the grain size of each ceramic filler contained in the dielectric layer **40**.

(62) The proportion of each material contained in the dielectric layers **30** and **40** is not limited to the proportion described above. However, the proportion of the crystallized glass contained in the dielectric layers **30** and **40** is desirably 80 (vol %) or more.

(63) In the dielectric layers **30** and **40**, the ceramic filler is not limited to alumina and zirconia. For example, the ceramic filler may contain quartz, titania, or the like instead of at least one of alumina or zirconia, or in addition to at least one of alumina or zirconia.

(64) The proportion of each of the crystallized glass, alumina, and zirconia contained in the dielectric layer **30** may be different from the proportion of each of the crystallized glass, alumina, and zirconia contained in the dielectric layer **40**.

(65) In the first embodiment, the wiring conductor is formed inside the wiring board. However, the wiring conductor may be formed on an outer surface of the wiring board.

(66) In the first embodiment, each of the dielectric layers **30** and **40** is made of a glass-ceramic and contains a crystallized glass. However, the glass contained in the dielectric layers **30** and **40** is not limited to a crystallized glass, and may be a non-crystallized glass. For example, both of the dielectric layers **30** and **40** may contain a non-crystallized glass. For example, the dielectric layer **30** may contain a crystallized glass, and the dielectric layer **40** may contain a non-crystallized glass. For example, the dielectric layer **40** may contain a crystallized glass, and the dielectric layer **30** may contain a non-crystallized glass. In this manner, according to the first embodiment, since the glass contained in the dielectric layers **30** and **40** is not limited to either a crystallized glass or a non-crystallized glass, the restriction of the structure of the wiring board **10** can be reduced.

Second Embodiment

(67) FIG. **8** is a longitudinal sectional view of a wiring board according to a second embodiment of the present disclosure, corresponding to the section taken along the line A-A in FIG. **1**. A wiring

board **10A** according to the second embodiment is different from the wiring board **10** according to the first embodiment in that a through via **50** penetrating the dielectric layers **30** and **40** is further provided. Hereinafter, differences from the first embodiment will be described. Common points with the wiring board **10** of the first embodiment are denoted by the same reference numerals, and description thereof will be omitted in principle and described as necessary.

(68) As shown in FIG. **8**, the through via **50** is formed in the dielectric layers **30** and **40**. The through via **50** is formed by filling a through hole (via) **50A** penetrating the dielectric layers **30** and **40** with a conductive paste **50B** and co-sintering with the ceramic. The conductive paste **50B** is made of, for example, copper. The through via **50** is exposed to the outside of the wiring board **10A** via a principal surface **40A** of the dielectric layer **40**. The through via **50** is exposed to the wiring conductor **20**. This causes the through via **50** to be electrically connected to the wiring conductor **20**. In the second embodiment, one end of the through via **50** in a penetrating direction is electrically connected to the wiring conductor **20**, and the other end of the through via **50** in the penetrating direction is exposed to the outside of the wiring board **10A**.

(69) A surface constituting the through hole **50A** is inclined with respect to a thickness direction TD of the wiring board **10A**. The diameter of the through hole **50A** decreases from the principal surface **40A** of the dielectric layer **40** toward the wiring conductor **20**. That is, the through via **50** has a tapered shape that reduces its diameter toward the wiring conductor **20**. In other words, the through via **50** has a truncated cone shape.

(70) According to the second embodiment, in the vicinity of the boundary between the through via **50** and the wiring conductor **20**, a decrease in conductivity of the wiring conductor **20** is prevented because of a reduction in roughness of the interface between the wiring conductor **20** and the dielectric layer **30**. Thus, since the roughness of the interface between a part of the through via **50** and the dielectric layer **30** is reduced, transmission loss between the wiring conductor **20** and the through via **50** can be reduced as compared with a configuration in which only the dielectric layer **40** is provided.

(71) Normally, in the through via **50** having a tapered shape, transmission loss is likely to occur in a part having a smaller diameter than a part having a larger diameter. According to the second embodiment, the part having a smaller diameter of the through via **50** where transmission loss is likely to occur is positioned near the boundary with the wiring conductor **20**. In the vicinity of the boundary, a decrease in conductivity of the wiring conductor **20** is prevented because of a reduction in roughness of the interface between the wiring conductor **20** and the dielectric layer **30**. Thus, transmission loss between the wiring conductor **20** and the through via **50** can be reduced.

(72) In the second embodiment, the through via **50** has a tapered shape that reduces its diameter toward the wiring conductor **20**, but the through via is not limited to such a shape. For example, the diameter of the through via **50** may be constant. That is, the through via **50** may have a cylindrical shape.

(73) In the second embodiment, the through via **50** is exposed to the outside of the wiring board **10A**. However, the through via **50** does not have to be exposed to the outside of the wiring board **10A**. For example, when the wiring board **10A** is a multilayer board, the through via **50** may be formed in a board constituting an inner layer of the wiring board **10A**.

(74) In the second embodiment, one through via **50** is formed in the wiring board **10A**, but the number of through via **50** is not limited to one and may be plural. In the second embodiment, the through via **50** is formed on the principal surface **40A** of the dielectric layer **40**, but the through via **50** may be formed on a surface other than the principal surface **40A**, for example, a principal surface **40B** of the dielectric layer **40**. The principal surface **40B** is the surface opposite to the principal surface **40A**. The through via **50** may be formed on a plurality of surfaces. For example, the through via **50** may be formed on both the principal surfaces **40A** and **40B**.

(75) By appropriately combining any embodiments among the various embodiments described above, the effects of the respective embodiments can be achieved.

(76) Although the present disclosure has been sufficiently described in connection with preferable embodiments with reference to the drawings as appropriate, various modifications and corrections are apparent to those skilled in the art. Such modifications and corrections should be understood to be included within the scope of the present disclosure according to the appended claims as long as they do not depart from the scope of the present disclosure.

EXPLANATION OF REFERENCE NUMBERS

(77) **10** wiring board **20** wiring conductor **30** dielectric layer (second dielectric layer) **40** dielectric layer (first dielectric layer) **50** through via

Claims

1. A wiring board comprising: a wiring conductor; a first dielectric layer around the wiring conductor and containing a first glass and a first ceramic filler; and a second dielectric layer interposed between the wiring conductor and the first dielectric layer, the second dielectric layer being in contact with the wiring conductor and the first dielectric layer, and the second dielectric layer containing a second glass and a second ceramic filler, wherein a sintering temperature of the second glass contained in the second dielectric layer is higher than a sintering temperature of the wiring conductor, and wherein a grain size of the second glass contained in the second dielectric layer is smaller than a grain size of the first glass contained in the first dielectric layer.
 2. The wiring board of claim 1, wherein a grain size of the second ceramic filler contained in the second dielectric layer is equal to or smaller than a grain size of the first ceramic filler contained in the first dielectric layer.
 3. The wiring board of claim 1, wherein the grain size of the first glass is 2.0 (μm) or less, and the grain size of the second glass is 1.0 (μm) or less.
 4. The wiring board of claim 1, further comprising a through via penetrating the first dielectric layer and the second dielectric layer and electrically connected to the wiring conductor.
 5. The wiring board of claim 4, wherein the through via has a tapered shape where the diameter thereof reduces in a direction toward the wiring conductor.
 6. The wiring board of claim 1, wherein the first ceramic filler and the second ceramic filler comprise alumina and zirconia.
 7. The wiring board of claim 6, wherein the first dielectric layer contains 98 (vol %) of the first glass, 1.5 (vol %) of the alumina, and 0.5 (vol %) of the zirconia.
 8. The wiring board of claim 7, wherein the second dielectric layer contains 98 (vol %) of the second glass, 1.5 (vol %) of the alumina, and 0.5 (vol %) of the zirconia.
 9. The wiring board of claim 6, wherein a grain size of each of the alumina and zirconia in the second ceramic filler contained in the second dielectric layer is equal to or smaller than a grain size of each of the alumina and zirconia in the first ceramic filler contained in the first dielectric layer.
 10. The wiring board of claim 1, wherein the sintering temperature of the crystallized glass contained in the second dielectric layer is 900 to 950 ($^{\circ}\text{C}.$), and the sintering temperature of the wiring conductor is 850 to 900 ($^{\circ}\text{C}.$).
 11. The wiring board of claim 1, wherein a thickness of the wiring conductor is larger than a thickness of the second dielectric layer.
 12. The wiring board of claim 11, wherein the thickness of the wiring conductor is 5 to 30 (μm), and the thickness of the second dielectric layer is 5 to 20 (μm).
 13. The wiring board of claim 1, wherein a thickness of the wiring conductor is equal to or less than a thickness of the second dielectric layer.
 14. The wiring board of claim 1, wherein the first glass contained in the first dielectric layer and the second glass contained in the second dielectric layer are a crystallized glass.
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